

$$30 + 5 + 16 + 0 + 2 + 16 + 8$$

$$\lim_{x \rightarrow \infty} \left(\frac{x^2}{x-1} + ax + b \right)$$

$$\lim_{x \rightarrow \infty} [1 + f(x)]^x$$

$$= \lim_{x \rightarrow \infty} e^{x \ln(1 + f(x))}$$

$$= e^{\lim_{x \rightarrow \infty} x \ln(1 + f(x))}$$

$$\frac{x^2 + (ax+b)(x-1)}{x-1}$$

$$x^2 + ax^2 - ax + bx - b$$

$$= \frac{x^2(a+1) + x(b-a) - b}{x-1}$$

$$2x(a+1) + b - a = 0$$

$$\begin{cases} a+1=0 & a=-1 \\ b-a=0 & b=-1 \end{cases}$$

$$x^2 + (-1-x)(x-1)$$

$$x^2 - (x+1)(x-1)$$

$$x^2 - (x^2 - 1)$$

$$f(x) = \frac{1}{x-1}$$

$$\lim_{x \rightarrow \infty} x \ln\left(1 + \frac{1}{x-1}\right)$$

$$x \rightarrow \infty$$

$$x = \frac{1}{t}$$

$$1 + \frac{1}{\frac{1}{1-t}} = 1 + \frac{t}{1-t}$$

$$\lim_{t \rightarrow 0} \frac{\ln\left(1 + \frac{t}{1-t}\right)}{t}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{t}{1-t}}{(1-t)t}$$

$$= 1$$

$$e \checkmark$$

$$2. \begin{cases} x = t^3 + t + 1 \\ y = t^3 - t + 1 \end{cases}$$

$$\lim_{n \rightarrow \infty} n \left[f\left(\frac{n+1}{n}\right) - f\left(\frac{n}{n}\right) \right]$$

$$\dot{n} = \frac{1}{x}$$

T6

$$\int_0^1 (2x+1)^2 e^{2x^2} dx$$

$$= \int_0^1 (4x^2 + 4x + 1) e^{2x^2} dx$$

$$= \int_0^1 4x^2 e^{2x^2} dx + \int_0^1 4x e^{2x^2} dx + \int_0^1 e^{2x^2} dx$$

$$\int_0^1 e^{2x^2} dx$$

$$2x^2 e^{2x^2} \Big|_0^1 - \int_0^1 2x^2 de^{2x^2}$$

$$= 2e^2 - \int_0^1 2x^2 de^{2x^2}$$

$$\int_0^1 x de^{2x^2}$$

$$+ x e^{2x^2} \Big|_0^1 - \int_0^1 x de^{2x^2}$$

$$3e^2 - \int_0^1 2x^2 de^{2x^2}$$

$$2x^2 = t \quad x: 0 \rightarrow 1 \quad t: 0 \rightarrow 2$$

$$\int_0^2 t de^t$$

$$= t e^t \Big|_0^2 - \int_0^2 e^t dt$$

$$t e^t \Big|_0^2 - e^t \Big|_0^2$$

$$2e^2 - 0 - (e^2 - 1)$$

$$e^2 + 1$$

$$2e^2 - 1 \checkmark$$

$$T7 \quad (x^2 + y^2)^2 = x^2 y^2 \quad \frac{1}{2}$$

$$T8 \quad z = f(x, y)$$

$$dz = \frac{(x+2y)dx + ydy}{(x+y)^2}$$

$$\frac{\partial z}{\partial x} = \frac{x+2y}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{y}{(x+y)^2} \quad 3$$

$$\lim_{x \rightarrow 0} \frac{f(1+4x) - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{4f'(1+4x)}{1}$$

$$= \lim_{x \rightarrow 0} 4f'(1+4x) = 4 \times \frac{1}{2} = 2$$

$$= 4f'(1)$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{3t^2 - 1}{3t^2 + 1}$$

$$= \frac{3+1}{3+1} = \frac{2}{4} = \frac{1}{2}$$

$$3. f(x) = \frac{1}{2-x^4}$$

$$\underbrace{f(x)(2-x^4)}_{uv} = 1$$

$$-4x^3 \quad -12x^2 \quad -16x \quad -16$$

$$C_n^0 (2-x^4) f^{(n)}(x) + C_n^1 (2-x^4)' f^{(n-1)}(x) + C_n^2 (2-x^4)^2 f^{(n-2)}(x) + C_n^3 (2-x^4)^3 f^{(n-3)}(x) + C_n^4 (2-x^4)^4 f^{(n-4)}(x) = 0$$

$$n=4, x=0$$

$$(2-x^4) f^{(4)}(0) + (-24) C_n^4 \frac{1}{2} = 0$$

$$2 f^{(4)}(0) - 12 \left(\frac{1}{2}\right) = 0$$

$$2 f^{(4)}(0) = 12$$

$$f^{(4)}(0) = 6 \quad \checkmark$$

$$14 \quad x = \sin t \quad \frac{dx}{dt} = \cos t$$

$$(1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} \quad y.t.$$

$$\cos^2 t$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{1}{\cos t} \frac{dy}{dt}$$

$$\frac{d^2y}{dx^2} = \frac{d\left(\frac{1}{\cos t} \frac{dy}{dt}\right)}{dx/dt}$$

$$= \frac{\frac{\sin t}{\cos^2 t} \frac{dy}{dt} + \frac{1}{\cos t} \frac{d^2y}{dt^2}}{\cos t}$$

$$\cos t \left[\frac{1}{\cos t} \frac{d^2y}{dt^2} + \frac{\sin t}{\cos^2 t} \frac{dy}{dt} \right] - \sin t \cdot \frac{1}{\cos t} \frac{dy}{dt}$$

$$\underline{d^2y}$$

$\frac{d}{dt} \sqrt{t^2+1}$ ✓

$$T5 \quad \int_2^{+\infty} \frac{dx}{x\sqrt{x-1}}$$

$$x-1=u$$

$$2 \int_1^{+\infty} \frac{d\sqrt{u}}{(u+1)}$$

$$2 \int_1^{+\infty} \frac{d\sqrt{u}}{(\sqrt{u})^2+1}$$

$$= 2 \arctan \sqrt{u} \Big|_1^{+\infty}$$

$$2\left(\frac{\pi}{2} - \frac{\pi}{4}\right)$$

$$2 \times \frac{\pi}{4} = \frac{\pi}{2} \quad \checkmark$$

Ex. 5

命题 1:

$f(x)$ 在 x_0 可导

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = k.$$

$$\lim_{x \rightarrow x_0} \frac{|f(x)|}{\frac{|f(x)| - |f(x_0)|}{x - x_0}}$$

$$= \lim_{x \rightarrow x_0} \frac{|f(x)| - |f(x_0)|}{x - x_0} \cdot \frac{x - x_0}{f(x) - f(x_0)} \cdot k$$

$$= k \lim_{x \rightarrow x_0} \frac{|f(x)| - |f(x_0)|}{f(x) - f(x_0)}$$

$$\begin{matrix} > 0 & k \\ < 0 & -k \end{matrix}$$

不可导

反例 $y = x$ ✓

命题 2:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$|f(x)|$ 在 x_0 处可.

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = k = 0.$$

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = 0. \quad \times$$

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$$\text{三. } f(x) = \int_0^x \sqrt{1+t} dt - \int_0^x \sqrt{1+t^2} dt$$

$$\begin{aligned} (1) f'(x) &= 2x\sqrt{1+x^2} - \sqrt{1+x^2} \\ &= (2x-1)\sqrt{1+x^2} \end{aligned}$$

当 $x > \frac{1}{2}$ 时 $f'(x) > 0$ $f(x) \uparrow$

当 $x < \frac{1}{2}$ 时 $f'(x) < 0$ $f(x) \downarrow$ ✓

$$\begin{aligned} (2) f_m &= f\left(\frac{1}{2}\right) = \int_0^{\frac{1}{2}} \sqrt{1+t} dt - \int_0^{\frac{1}{2}} \sqrt{1+t^2} dt \\ &= \left[\frac{2}{3}(1+t)^{\frac{3}{2}} \right] \Big|_0^{\frac{1}{2}} - \left[\frac{x}{2}\sqrt{1+x^2} - \frac{1}{2}\ln(x+\sqrt{1+x^2}) \right] \Big|_0^{\frac{1}{2}} \\ &= \frac{\sqrt{5}+6}{24} - \frac{1}{2}\ln\frac{1+\sqrt{5}}{2} \quad \checkmark \end{aligned}$$

$$(3) f(x) = \frac{2}{3}[(1+x^2)^{\frac{3}{2}} - 1] - \frac{x}{2}\sqrt{1+x^2} - \frac{1}{2}\ln(x+\sqrt{1+x^2})$$

$$F(x) = f(x) - a \quad F(x)_m = -a$$

① $f_m > 0$, $a < f_m$ 无实根 ② $f_m = 0$ 有一根 ③ $f(-\infty) = f(+\infty)$

$$\text{四. } (1) \lim_{n \rightarrow \infty} \frac{1}{n} \int_0^1 \frac{n^x}{x+1} dx$$

$$\lim_{n \rightarrow \infty} \frac{\int_0^1 \frac{n^x}{x+1} dx}{n} = \lim_{n \rightarrow \infty} \frac{\int_0^1 n^x d\ln(1+x)}{n} \quad \times$$

$$(2) \lim_{n \rightarrow \infty} n \int_0^1 \frac{x^{n+2.25} - x^n}{\ln x} dx$$

$$\int_0^1 \frac{x^{n+2.25} - x^n}{\ln x} dx$$

$$\int_0^1 \frac{x^n (x^{2.25} - 1)}{\ln x} dx \quad \times$$

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} \int_0^1 \frac{x^{2.25} - 1}{\ln x} dx^{n+1}$$

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$$\text{五. } f(x,y) = x^3 + axy - y^3$$

$$\frac{\partial f}{\partial x} = 3x^2 + ay = 0$$

$$\frac{\partial f}{\partial y} = ax - 3y^2 = 0$$

$$\text{则 } \frac{\partial^2 f}{\partial x^2} = 6x$$

$$\int_0^x \sqrt{1+t^2} dt$$

$$\frac{2}{3}[(1+x^2)^{\frac{3}{2}} - 1]$$

$$\int_0^x \sqrt{1+t^2} dt$$

$$t\sqrt{1+t^2} \Big|_0^x - \int_0^x \frac{t^2}{\sqrt{1+t^2}} dt$$

$$\int_0^x \frac{t^2}{\sqrt{1+t^2}} dt$$

$$t = \tan \theta \quad t: 0 \rightarrow x \quad \theta: 0 \rightarrow \arctan x$$

$$\int_0^{\arctan x} \frac{\tan^2 \theta}{\sec \theta} \sec^2 \theta d\theta$$

$$= \int_0^{\arctan x} (1 - \sec^2 \theta) \sec \theta d\theta$$

$$I = \int_0^{\arctan x} \sec \theta d\theta - I + x\sqrt{1+x^2}$$

$$I = \frac{1}{2} x\sqrt{1+x^2} + \frac{1}{2} \ln(x+\sqrt{1+x^2})$$

$$\int_0^{\arctan x} \sec \theta d\theta$$

$$\int \sec \theta d\theta = \int \frac{\sec \theta (\sec \theta + \tan \theta)}{1(\sec \theta + \tan \theta)}$$

$$= \ln(\sec \theta + \tan \theta) \Big|_0^{\arctan x}$$

$$\ln(x + \sqrt{1+x^2})$$

$$C \frac{\partial^2 f}{\partial y^2} = -6y$$

$$13 \frac{\partial^2 f}{\partial x \partial y} = a$$

$$AC - B^2 = (a^2 + 36xy)$$

$$\begin{cases} 3x^2 + ay = 0 \\ ax - 3y^2 = 0 \end{cases}$$

$$(0,0) \quad \left(\frac{a}{3}, -\frac{a}{3}\right)$$

$$① a=0, (0,0)$$

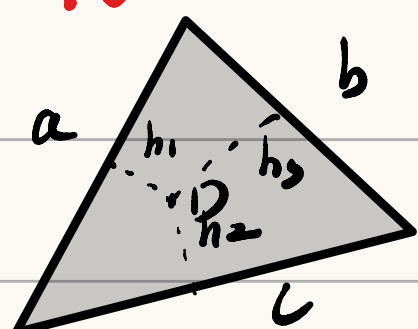
$$② a \neq 0 \quad (0,0) \quad AC - B^2 = -a^2 < 0 \text{ 不是}$$

$$\left(\frac{a}{3}, -\frac{a}{3}\right) \quad AC - B^2 = 3a^2 > 0$$

$$1^\circ a > 0 \text{ 时}, f_{\min}\left(\frac{a}{3}, -\frac{a}{3}\right) = -\frac{a^3}{27}$$

$$a < 0 \text{ 时}, f_{\max}\left(\frac{a}{3}, -\frac{a}{3}\right) = -\frac{a^3}{27} \checkmark$$

六. 16.



$$t = xyz$$

$$S = \frac{1}{2}ah_1 + \frac{1}{2}bh_2 + \frac{1}{2}ch_3$$

$$= \frac{1}{2}(xh_1 + yh_2 + zh_3)$$

$$xh_1 + yh_2 + zh_3 = 2S = 2$$

$$L(x, y, z, \lambda)$$

$$= xyz - \lambda(ax + by + cz - 2A)$$

$$L'_x = yz - a$$

$$L'_y = xz - b$$

$$L'_z = xy - c$$

$$L'_\lambda = ax + by + cz - 2A$$

$$\text{令 } L'_x, L'_y, L'_z, L'_\lambda \text{ 均为 } 0$$

$$a = yz \quad b = xz \quad c = xy$$

$$xyz + xyz + xyz = 2A$$

$$3xyz = 2A$$

$$\begin{cases} x = \frac{2A}{3yz} = \frac{2A}{3a} \\ y = \frac{2A}{3xz} = \frac{2A}{3b} \\ z = \frac{2A}{3xy} = \frac{2A}{3c} \end{cases}$$

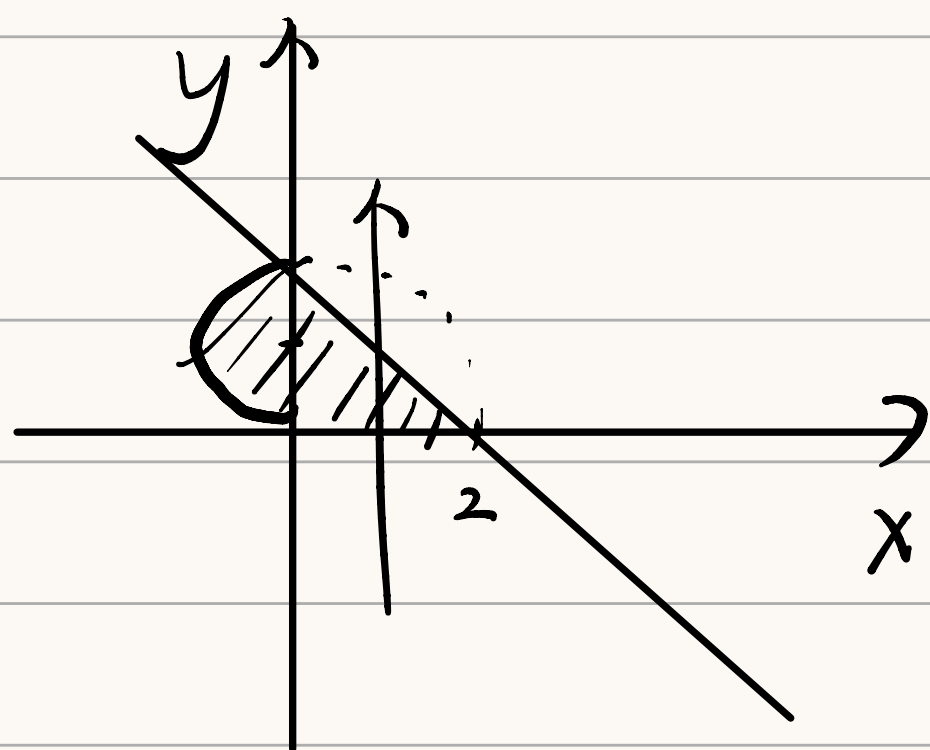
$$xyz = \frac{2A}{3}$$

$$i. S_{max} = \frac{8\pi}{27abc} \checkmark$$

$$t. 8 \quad x = -\sqrt{2y-y^2}$$

$$y = 2-x$$

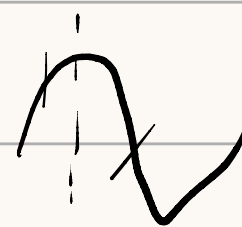
$$I = \iint_D \frac{y^2}{x^2+y^2} dx dy$$



$$x^2 + y^2 = 2y$$

$$y^2 - 2y + 1 + x^2 = 1$$

$$(y-1)^2 + x^2 = 1$$



$$x = \cos \theta$$

$$y = \sin \theta$$

$$I_1 = \int_{\frac{\pi}{2}}^{\pi} d\theta \int_0^{2\sin\theta} \frac{\sin^2\theta}{1} r dr$$

$$= 2 \int_{\frac{\pi}{2}}^{\pi} \sin^4 \theta d\theta$$

$$= 2 \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta$$

$$= 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{3}{8}\pi \checkmark$$

I_2 X

$$11. \quad |f(y)| \leq k|f(x)|$$

$$f'(x) = f(x) \quad (a,b)$$

$$g(x) = f'(x) - f(x)$$

$$g'(x) = f''(x) - f'(x)$$

$$|f(y)| = k|f(x)|$$

$$(f(y))^2 \leq (f(x))^2$$

$$y > x \quad f(y) < f(x)$$

$$f(b) - f(a) = f'(\xi)(b-a)$$

$$f(y) - f(x) = f'(\xi)(y-x)$$

$$f'(\xi) < 0$$

补题:

$$T7. (x^2+y^2)^2 = x^2 - y^2 \quad (x > 0)$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$$

$$(r^2)^2 = r^2(\cos^2 \theta - \sin^2 \theta) \quad \text{极坐标面积}$$

$$r^2 = \cos 2\theta \quad x > 0 \quad \theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$$

$$A = \frac{1}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \cos 2\theta d\theta$$

$$= \frac{1}{2} \sin 2\theta \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2}$$

$$T8. z = f(x, y) \quad dz = \frac{(x+2y)dx + ydy}{(x+y)^2} \quad \text{求 } f(2, 1) - f(1, 2)$$

$$\frac{\partial z}{\partial x} = \frac{x+2y}{(x+y)^2} = \frac{1}{x+y} + \frac{y}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{y}{(x+y)^2}$$

$$z = \int \left(\frac{1}{x+y} + \frac{y}{(x+y)^2} \right) dx$$

$$= \int \frac{1}{x+y} dx + \int \frac{y}{(x+y)^2} dx$$

$$= \ln|x+y| + y \int (x+y)^{-2} d(x+y)$$

$$= \ln(x+y) - \frac{y}{x+y} + g(y)$$

$$\frac{\partial z}{\partial y} = \frac{1}{x+y} - \frac{x+y-y}{(x+y)^2} + g'(y)$$

$$= \frac{y}{(x+y)^2} + g'(y)$$

$$\therefore g'(y) = 0 \quad \therefore g(y) = C$$

$$f(x,y) = \ln(x+y) - \frac{y}{x+y} + C$$

$$\therefore f(2,1) - f(1,2) = -\frac{1}{1} + C - \left[-\frac{2}{1+2} + C\right] = 3$$

二. 命题2:

$f(x)$ 在 x_0 处连续且 $|f(x)|$ 在 x_0 处可导, 则 $f(x)$ 在 x_0 处可导

证明:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

若 $f(x_0) > 0$, 则存在 $x \in U(x_0)$, s.t. $x \in U(x_0), f(x) > 0$

$\therefore |f(x)|$ 在 x_0 处可导

$$\therefore \lim_{x \rightarrow x_0} \frac{|f(x)| - |f(x_0)|}{x - x_0}$$

$$= \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \text{ 存在}$$

$\therefore f(x)$ 在 x_0 处可导

若 $f(x_0) < 0$, 则存在 $x \in U(x_0)$, s.t. $x \in U(x_0), f(x) < 0$

$\therefore |f(x)|$ 在 x_0 处可导

$$\therefore \lim_{x \rightarrow x_0} \frac{|f(x)| - |f(x_0)|}{x - x_0}$$

$$= - \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

存在.

∴ $f(x)$ 在 x_0 处可导

证 $f(x_0) = 0$ 时,

$$\lim_{x \rightarrow x_0} \frac{|f(x)| - |f(x_0)|}{x - x_0}$$

$$= \lim_{x \rightarrow x_0} \frac{|f(x)|}{x - x_0} \text{ 存在}$$

$$\text{又 } \lim_{x \rightarrow x_0} f(x) = f(x_0) = 0$$

$$\therefore \lim_{x \rightarrow x_0} \frac{f(x)}{x - x_0} \text{ 存在}$$

即 $f(x)$ 在 x_0 处可导

$$\text{T四. (1)} \lim_{n \rightarrow \infty} \frac{1}{n} \int_0^1 \frac{x^n}{x+1} dx \quad (2) \lim_{n \rightarrow \infty} n \int_0^1 \frac{x^{n+2025} - x^n}{\ln x} dx$$

(1) 由积分中值定理得:

$$\frac{1}{n} \int_0^1 \frac{x^n}{x+1} dx = \frac{1}{n} \cdot \frac{1}{\xi+1} \quad (0 < \xi < 1)$$

$$\text{则 } \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \frac{1}{\xi+1}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n+1} \cdot \frac{1}{1-\xi} = 0$$

$$(2) \because \frac{x^a - x^b}{\ln x} = \int_b^a x^t dt$$

$$\therefore \int_0^1 \frac{x^a - x^b}{\ln x} dx = \int_0^1 \int_b^a x^t dt dx$$

$$= \int_b^a \int_0^1 x^t dx dt$$

$$= \int_b^a \frac{1}{t+1} dt$$

$$= \ln \frac{a+1}{b+1}$$

$$\therefore \int_0^1 \frac{x^{n+2025} - x^n}{\ln x} dx$$

$$= \ln \frac{n+2026}{n+1}$$

$$\therefore \lim_{n \rightarrow \infty} n \left| \frac{n+2026}{n+1} \right|$$

$$= 2025$$

七. $a=0$ 时, 唯一驻点 $(0,0)$

由于在 $(0,0)$ 任意邻域内 $f(x,y)$ 可正可负
故该点不是极值点

七. $x+y=2$

$$r = \frac{2}{\sin\theta + \cos\theta}, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

$$I_1 = \int_0^{\frac{\pi}{2}} d\theta \int_0^{\frac{2}{\sin\theta + \cos\theta}} \sin^2\theta r dr$$

$$= \int_0^{\frac{\pi}{2}} \frac{2\sin^2\theta}{(\sin\theta + \cos\theta)^2} d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{2\tan^2\theta}{(1+\tan\theta)^2} d\theta$$

$$= \int_0^{+\infty} \frac{2t^2}{(1+t)^2} \frac{1}{1+t^2} dt$$

$$\frac{2t^2}{(1+t)^2(1+t^2)} = \frac{A}{(1+t)^2} + \frac{B}{1+t} + \frac{Ct+D}{1+t^2}$$

$$A(1+t^2) + B(1+t)(1+t^2) + (1+t)^2(Ct+D)$$

$$At^2 + A + B + Bt^2 + Bt + Bt^3 + Ct^3 + 2Ct^2 + Ct + Dt^2 + 2Dt + D$$

$$\begin{cases} A+B+D+2C=2 \\ B+C=0 \\ B+C+D=0 \\ A+B+D=0 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \\ C=1 \\ D=0 \end{cases}$$

11. $f(x) \in [a, b]$ $x \in [a, b]$, $y \in [a, b]$

$$|f(y)| \leq k|f(x)| \quad k \in (0, 1)$$

证明: $f'(x) = f(x)$ 在 (a, b) 内至少有一实根

证: 设 $g(x) = e^x f(x)$, 则 $g'(x) = e^x (f'(x) - f(x))$

(1) 若存在 $x_0 \in [a, b]$ 使 $f(x_0) = 0$, $g(x_0) = 0$

存在 $x_1 \in [a, b]$, $x_1 \neq x_0$, 使:

$$|f(x_1)| \leq k|f(x_0)| = 0$$

即 $f(x_1) = 0$ 则 $g(x_1) = 0$

由罗尔定理, x_0, x_1 之间存在 ξ , 使:

$$g'(\xi) = e^\xi (f'(\xi) - f(\xi)) = 0$$

即 $f'(x) = f(x)$ 在 (a, b) 内至少有一个实根

(2) 若 $\forall x \in [a, b]$ $f(x) \neq 0$,

设 $f(x) > 0$, $\forall x \in [a, b]$, 则

$$g(x) > 0, \forall x \in [a, b]$$

假设任给 $x \in [a, b]$, 皆随 $g'(x) \neq 0$

则 $g'(x)$ 恒为正或恒为负

设 $g'(x) > 0$

则 $g(x)$ 在 $[a, b]$ 上 ↑

对 $x = a$, 存在 $y \in [a, b]$, $y > a$, 使得:

$$|f(y)| \leq k|f(a)|, \text{ 且 } 0 < e^{a-y} < 1$$

$$\because f(x) = g(x) e^x$$

$$\therefore |g(y)| e^y \leq k |g(a)| e^a, \text{ 又 } k \in (0, 1)$$

$$\therefore |g(y)| \leq k |g(a)| e^{a-y} \leq |g(a)|$$

又 $g(x)$ 在 $[a, b]$ 上 ↑

$$\therefore g(y) > g(a) > 0$$

则 $|g(y)| > |g(a)|$. 矛盾, 假设不成立.

则 $g'(x)$ 不可能恒为正

故 (a, b) 内至少一点 ξ , 使得:

$$g'(\xi) = e^\xi (f'(\xi) - f(\xi)) = 0$$

即 $f'(x) = f(x)$ 在 (a, b) 内至少有一实根

